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Milano

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SQL Server Data Virtualization con PolyBase

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Chi sono?

- Gianluca Hotz | @glhotz | ghotz@ugiss.org
- Consulente indipendente
  - 20+ anni su SQL Server (dalla 4.21 nel 1996)
  - Modellazione e sviluppo database, dimensionamento e amministrazione database server, aggiornamenti e migrazioni, performance tuning
- Community
  - 20+ anni Microsoft MVP SQL Server/Data Platform (dal 1998)
  - VMware Experts SQL Server
  - Fondatore e president UGISS (PASS Chapter)
  - Co-organizzatore DAMAG Meetup Community

Agenda

- Introduzione
- Tecnologie per la virtualizzazione
  - Linked Server
  - PolyBase
- Scenari avanzati
  - Scale-Out
  - SQL Server Big Data Cluster (BDC)
- Altre piattaforme

Problema

- Sistemi DSS/OLAP/DWH necessitano di processi
  - ETL (Extract, Transform, Load)
  - ELT (Extract, Load, Transfrom)
- Causano «Data Movement»!
- Mitigata da virtualizzazione accesso ai dati

Data Movement

- Costi
  - Costi storage duplicati
  - Sforzo per costruire/mantenere pipeline dati
- Velocità
  - Tempo integrazione nuovi dati -> sfuggono opportunità!
  - Latenza disponibilità dati -> sfuggono opportunità
- Sicurezza
  - Superfice di attacco più ampia
  - Potenziale inconsistenza politiche di sicurezza
- Qualità
  - Pipeline dati possono introdurre problemi di qualità…
- Compliance
  - Maggiori problemi di governance

Virtualizzazione

- https://en.wikipedia.org/wiki/Data_virtualization
  - “Unlike the traditional extract, transform, load (ETL) process, the data remains in place, and real-time access is given to the source system for the data. This reduces the risk of data errors, of the workload moving data around that may never be used, and it does not attempt to impose a single data model on the data”
  - Dati rimangono dove sono
  - Viene dato accesso in tempo reale al dato
  - Non si cerca di imporre un solo modello…

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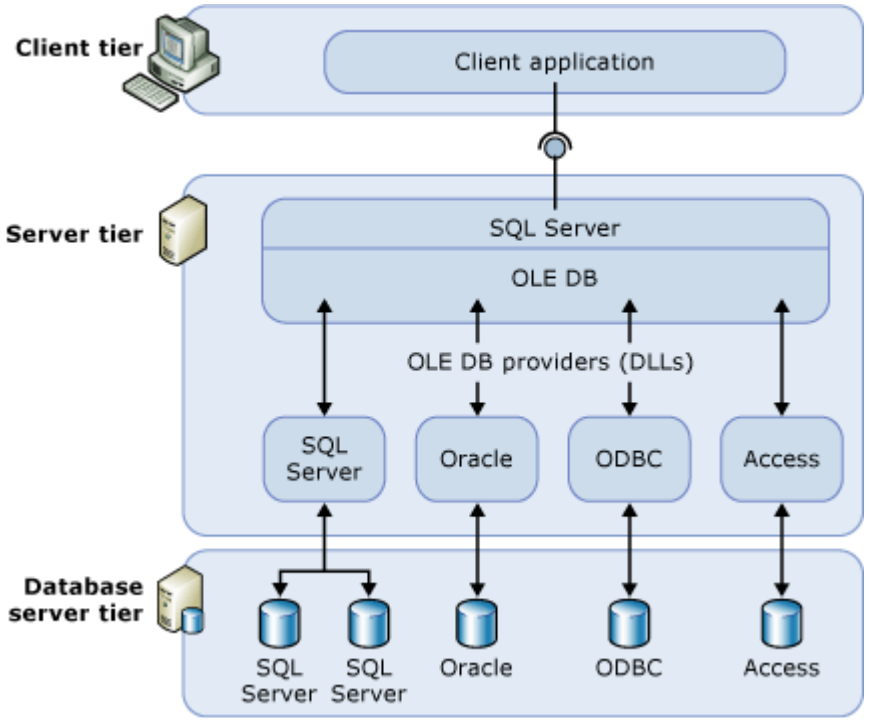
Virtualizzazione e SQL Server

- Open Data Service Gateway (< SQL Server 7.0!!)
- Linked Servers
- PolyBase

Linked Servers

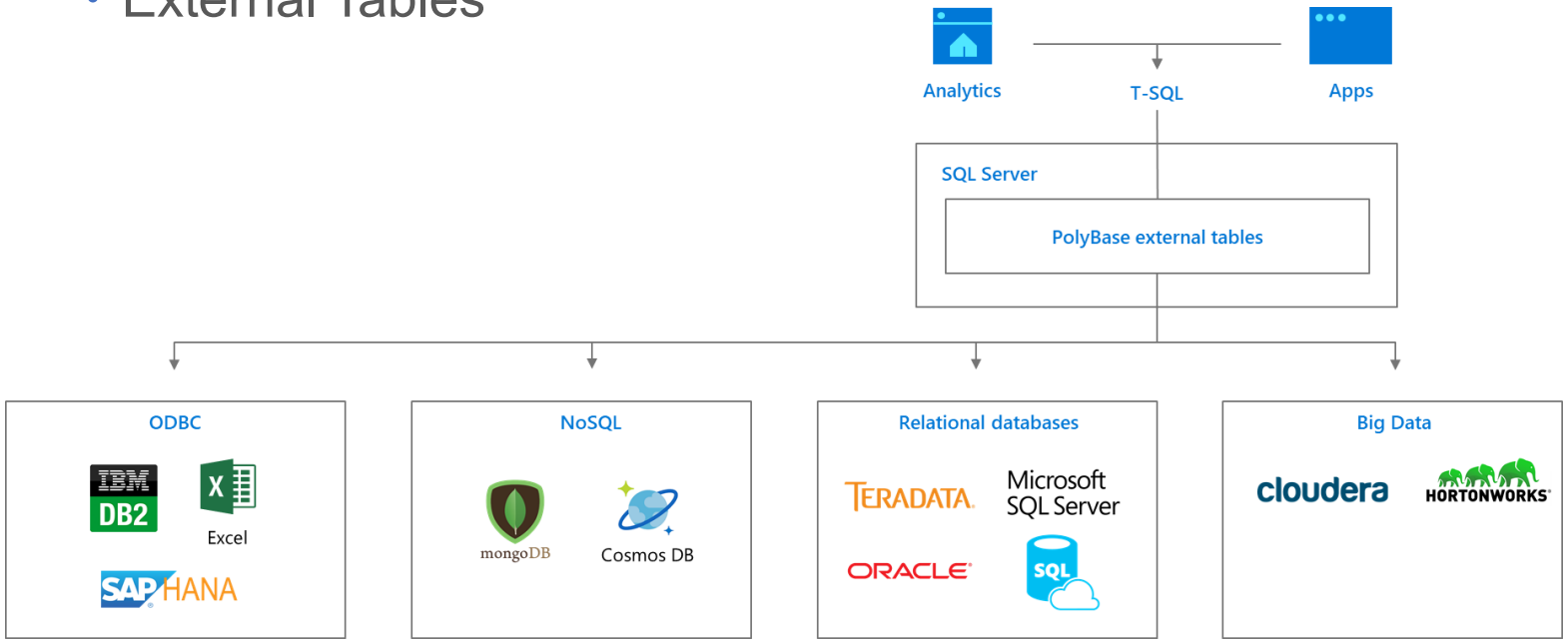
- Componenti
  - OLE DB provider
  - OLE DB data source
- Utilizzo
  - Accesso a fonti dati esterne eterogenee
  - Query e transazioni distribuite
- Supporto
  - SQL Server
  - Azure SQL Database Managed Instance

DEMO:
Linked Servers



PolyBase

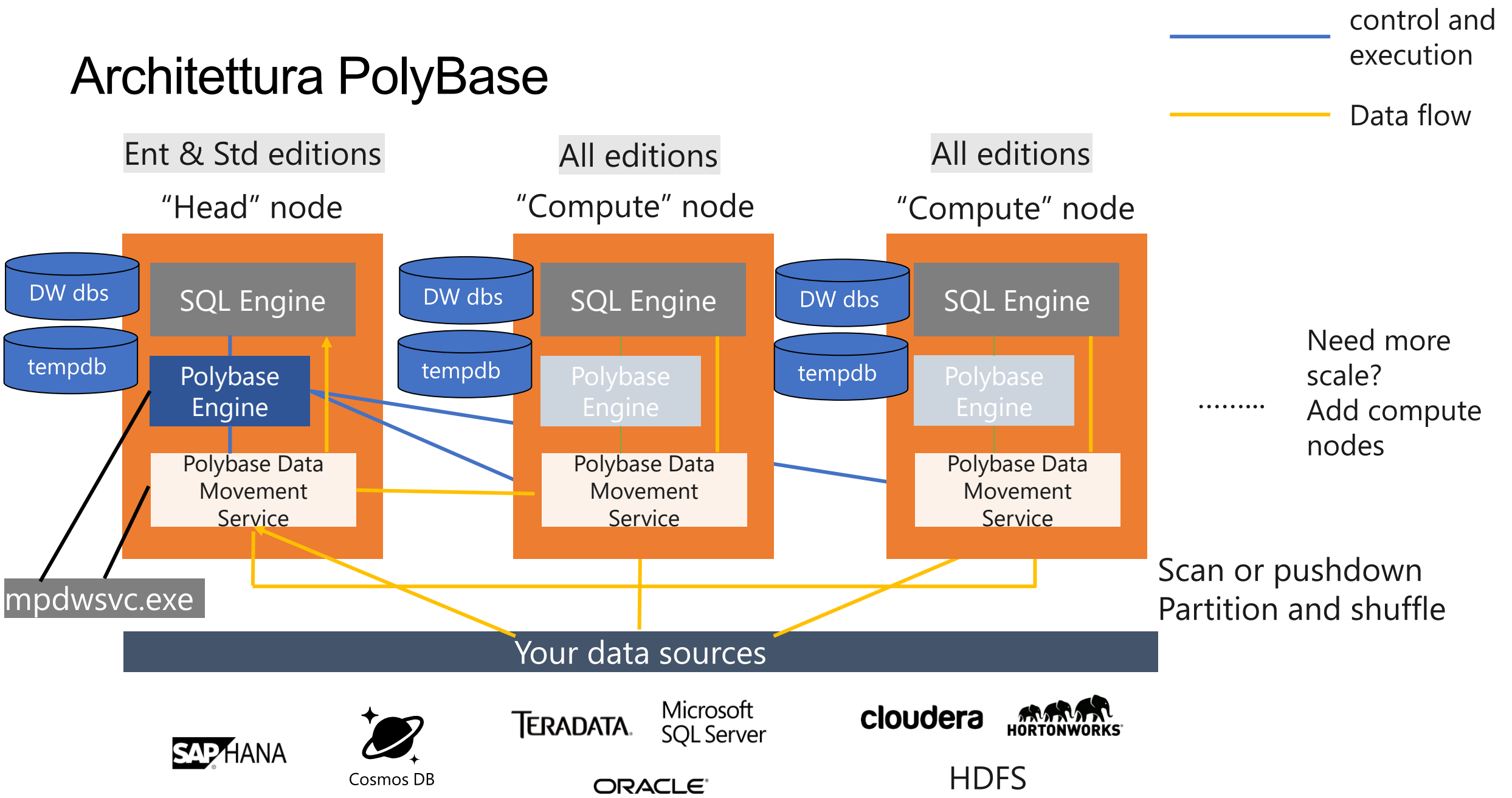
- Distributed Compute Engine
  - Integrato con SQL Server
  - Combina fonti eterogenee tramite query distribuite
- Componenti Virtualizzazione
  - External Data Source
  - External Tables



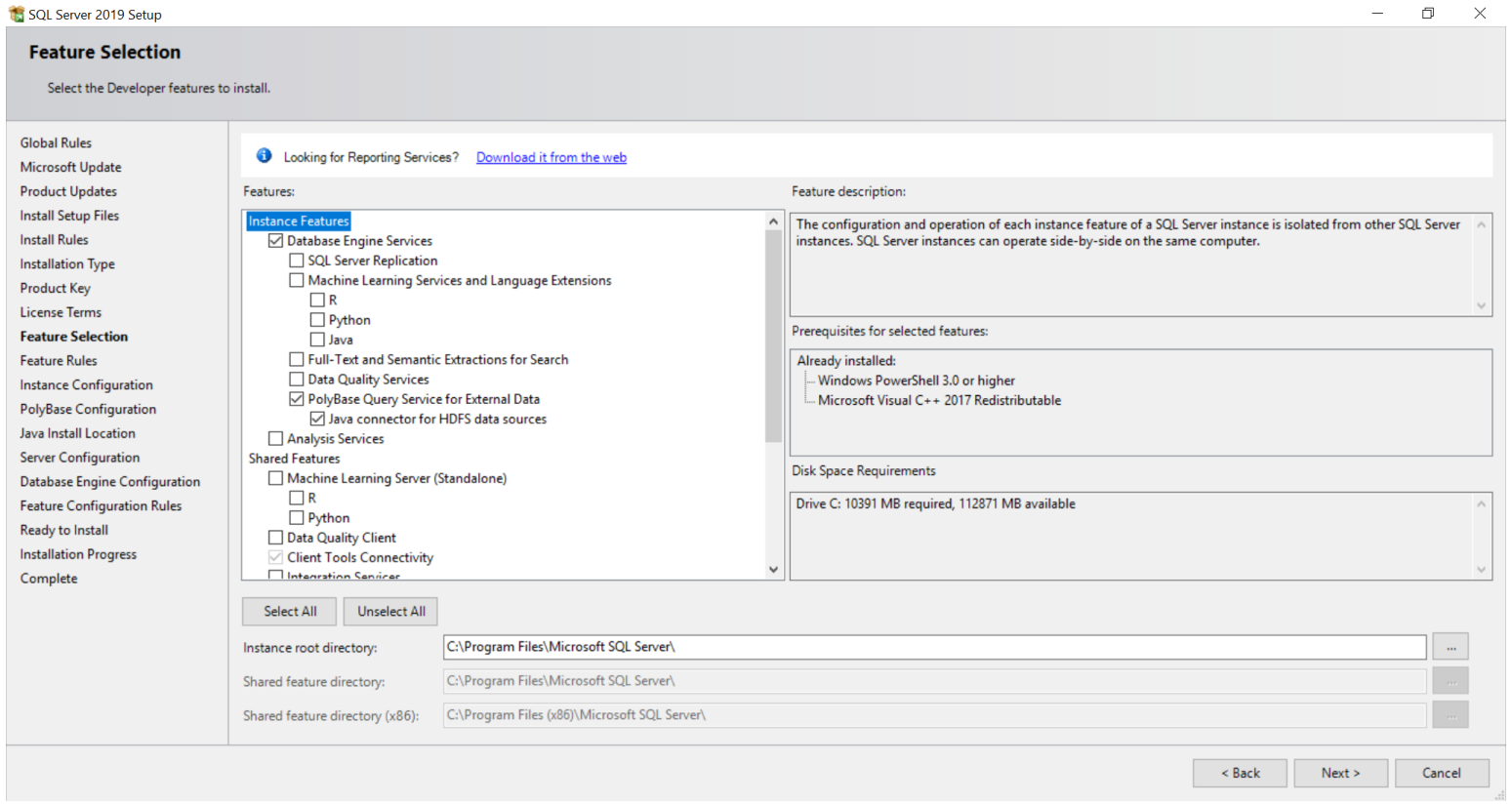
Supporto PolyBase

- SQL Server 2016+
  - Azure Blob Storage, Hadoop (Cloudera or Hortonworks)
- (SQL Server 2017+)
  - Operazioni BULK INSERT/OPENROWSET Blob Storage
- SQL Server 2019
  - SQL Server (all flavors), Oracle, Teradata, MongoDB, CosmosDB (Mongo API)
  - ODBC Generic Type (e.g. DB2, SAP Hana, Excel, …)

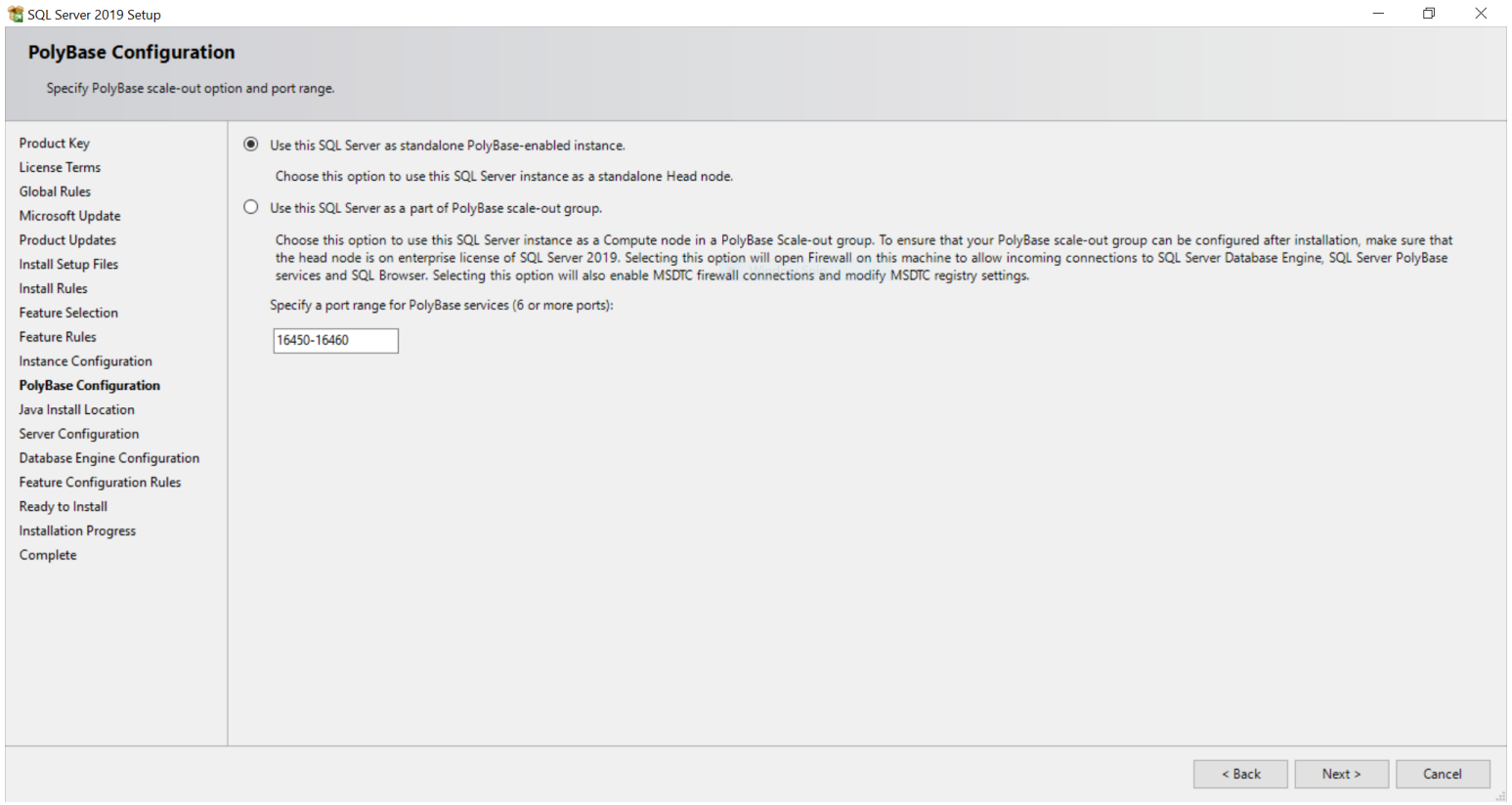
Architettura PolyBase



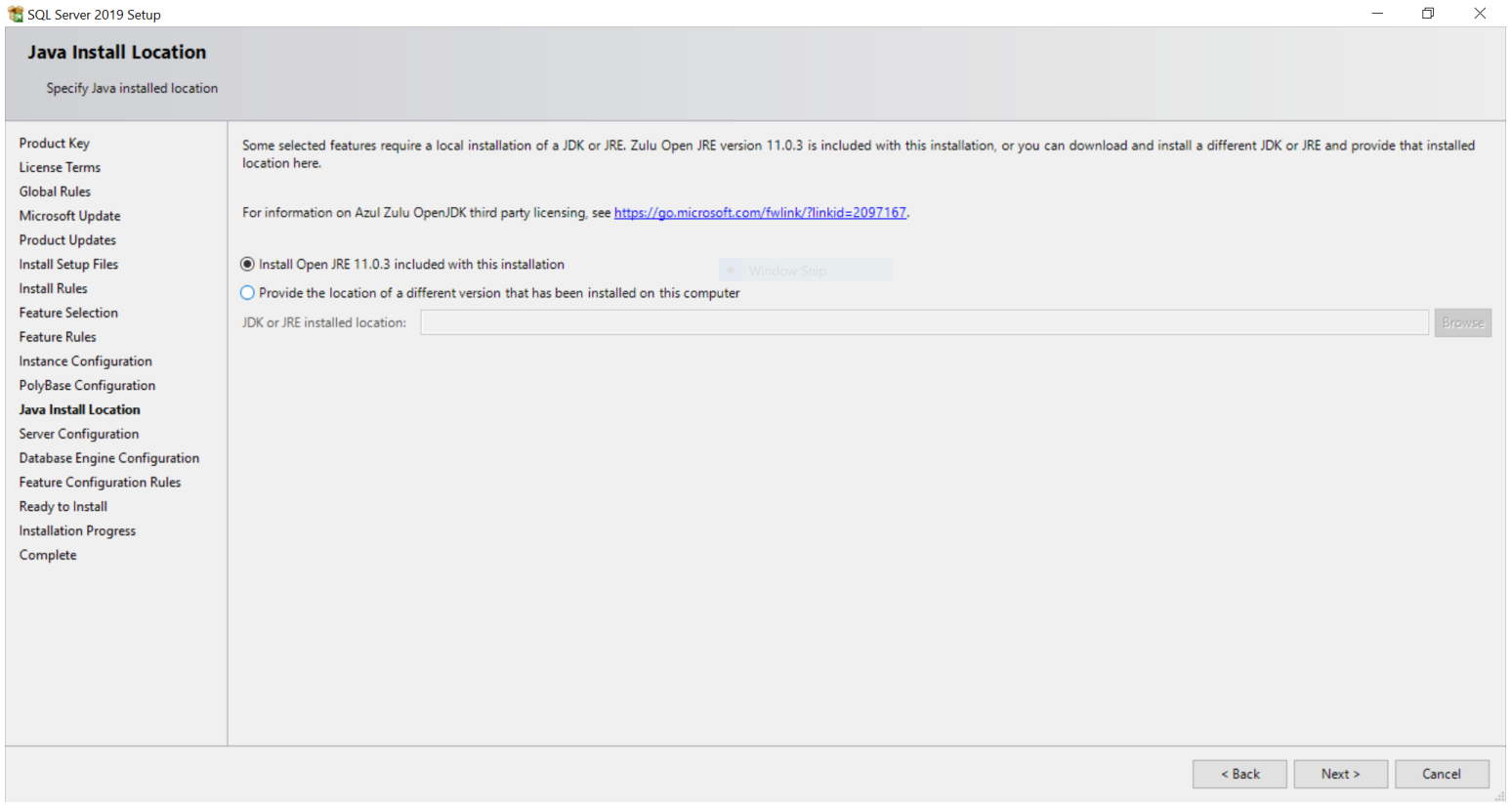
Setup: Feature Selection



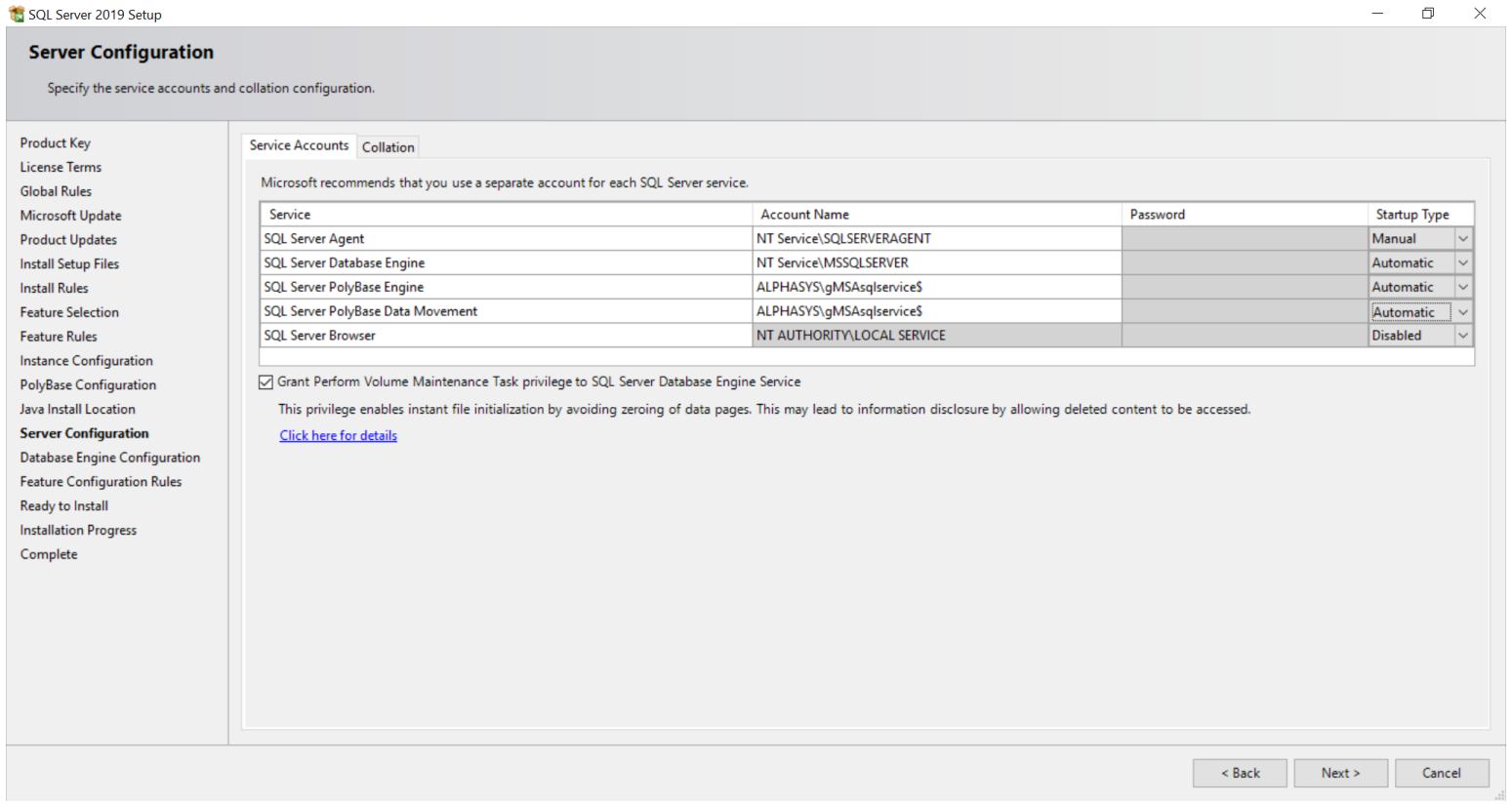
Setup: PolyBase Configuration



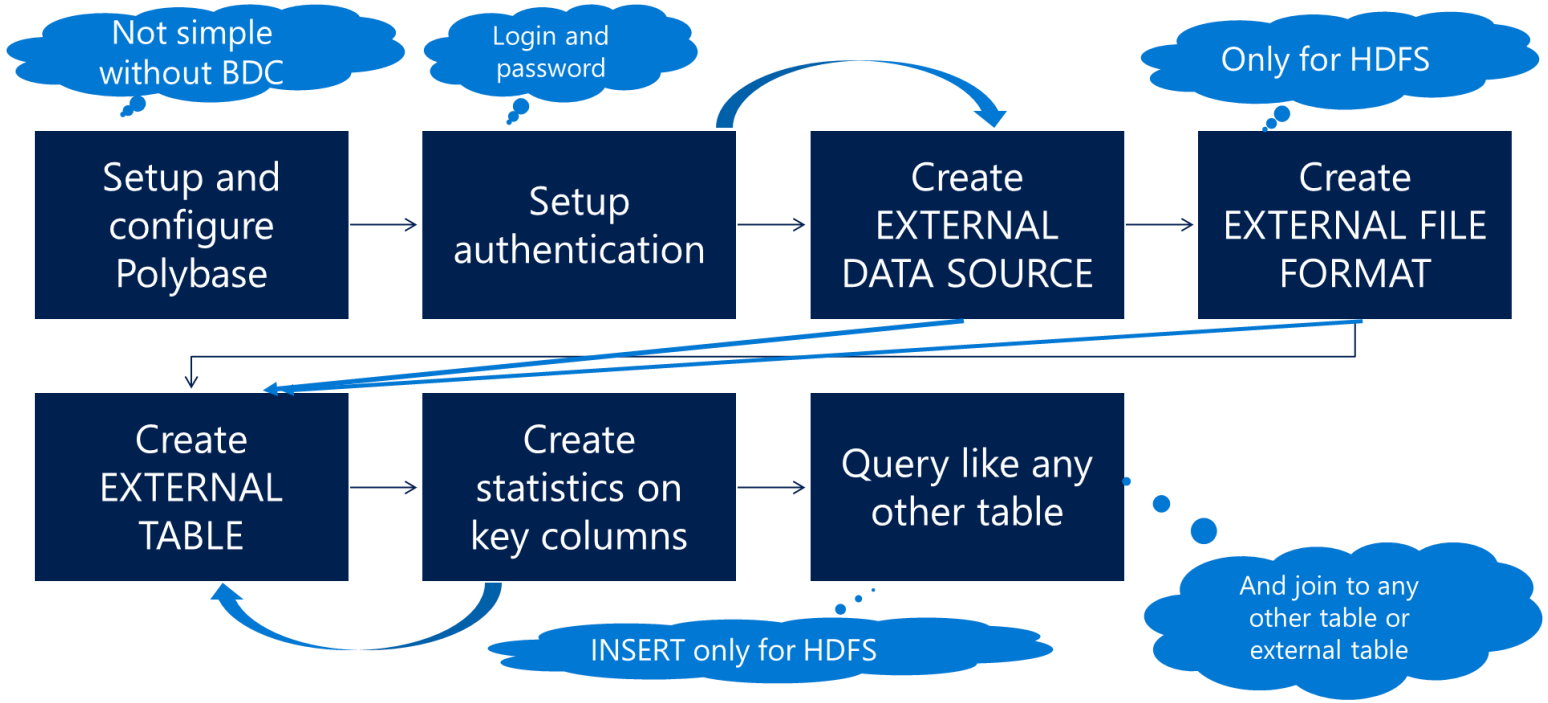
Java Install Location



Setup: Service Accounts



DEMO: PolyBase



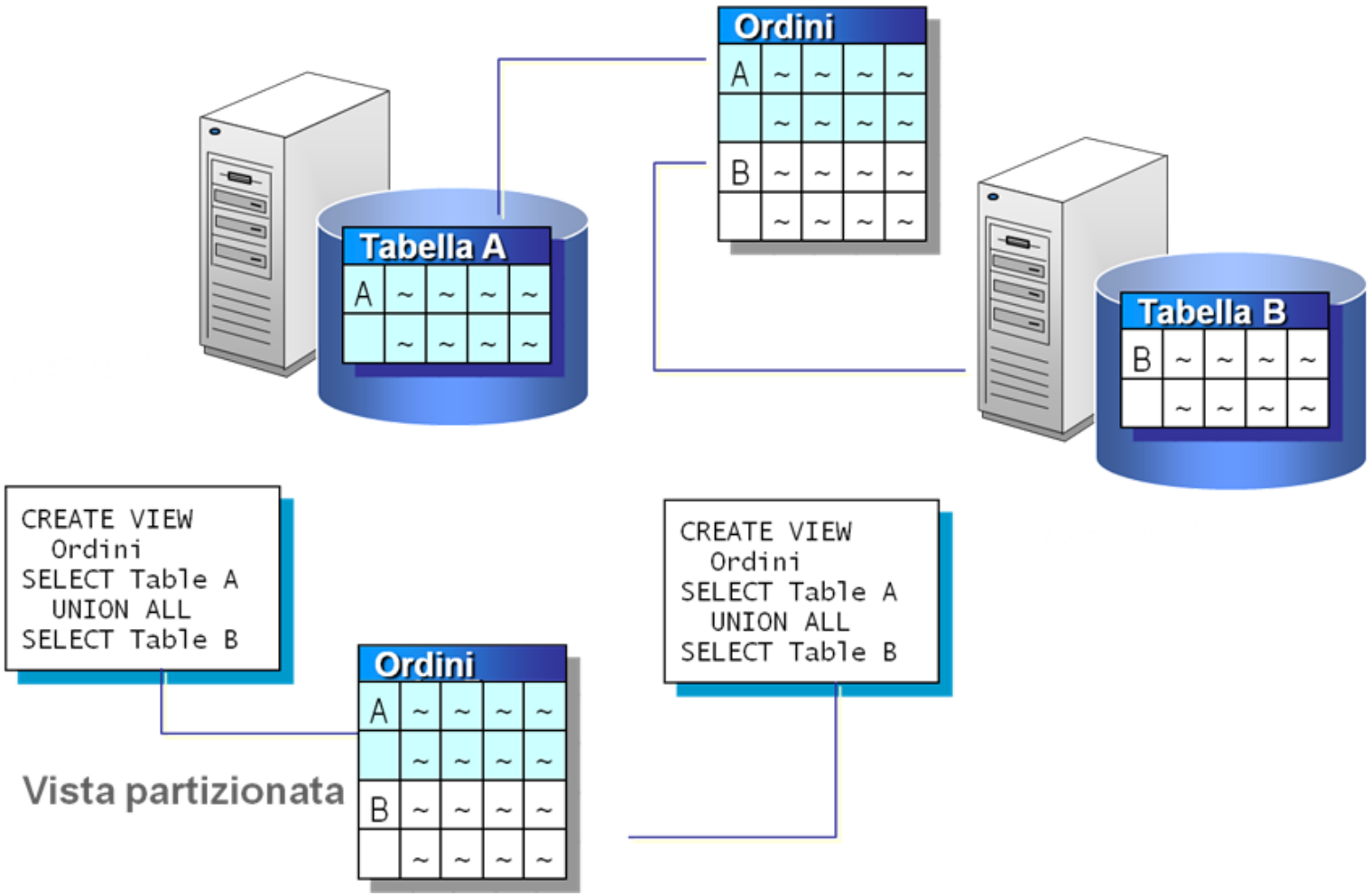
PolyBase con Azure SQL Database

- Bug nella versione 2019 RTM
  - https://github.com/MicrosoftDocs/sql-docs/issues/3727
  - Bisogna specificare database nel data source
  - `CONNECTION_OPTIONS = 'Database=pbdemo'`
  - Fix non documentato nella GDR
    - https://support.microsoft.com/en-us/help/4517790

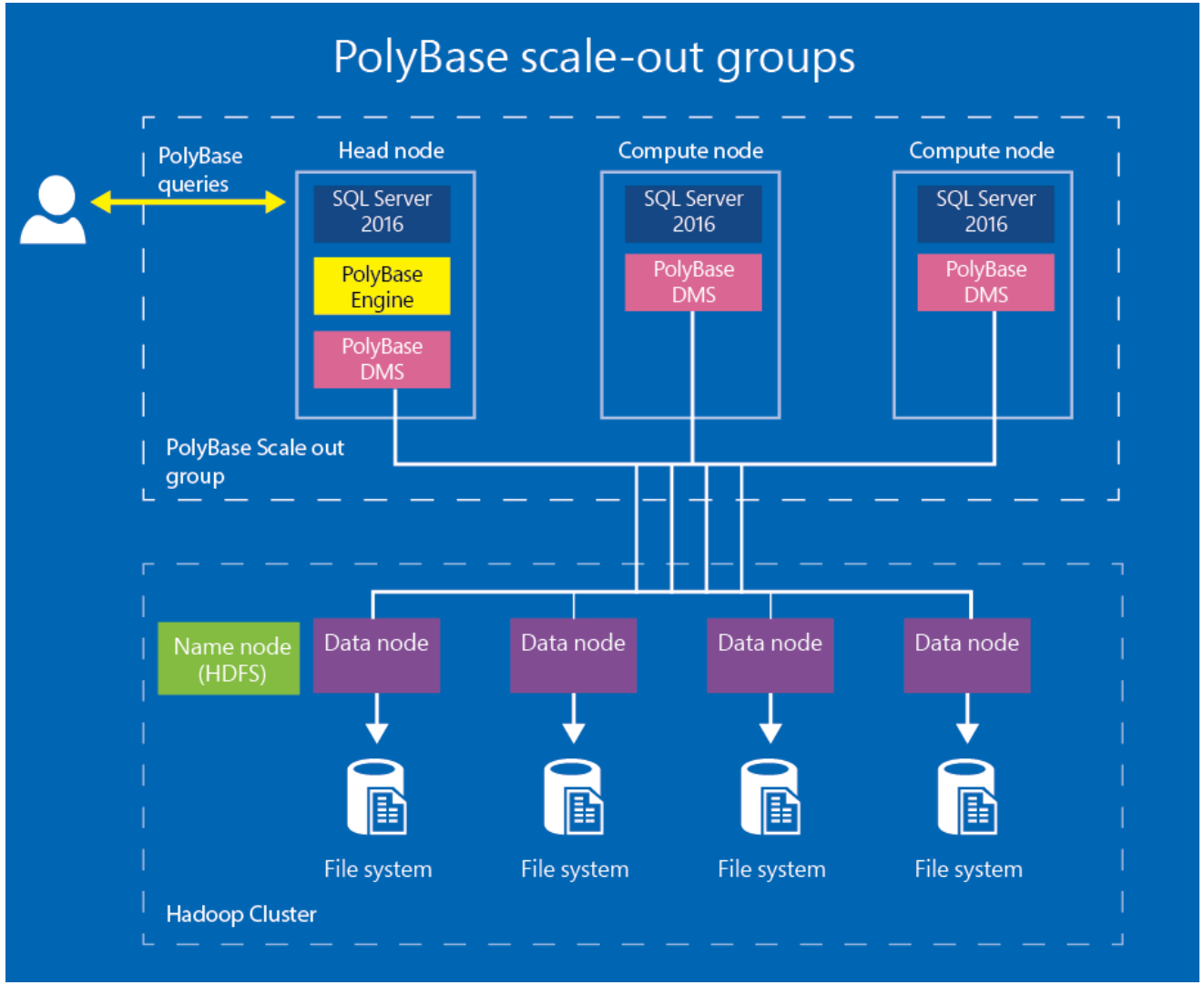
Scale-Out con Linked Server

- Distributed Partioned Views
  - Aka Federated Database Server in SQL Server 2000
- Partizionamento orizzontale tramite viste
  - Tabelle con partizioni dati distribuite su più istanze
  - Vincoli CHECK identificano la partizione
  - Viste su tutti i nodi con SELECT * […] UNION ALL
- Query Processor riconosce lo scenario
  - Distribuisce il carico eseguendo query remote/locali
  - Predicato deve operare su colonna del vincolo CHECK
- Disponibili ancora solo con edizione Enterprise
- Funzionalità poco documentata…

DEMO: Viste partizionate distribuite



DEMO: PolyBase Scale-Out Groups



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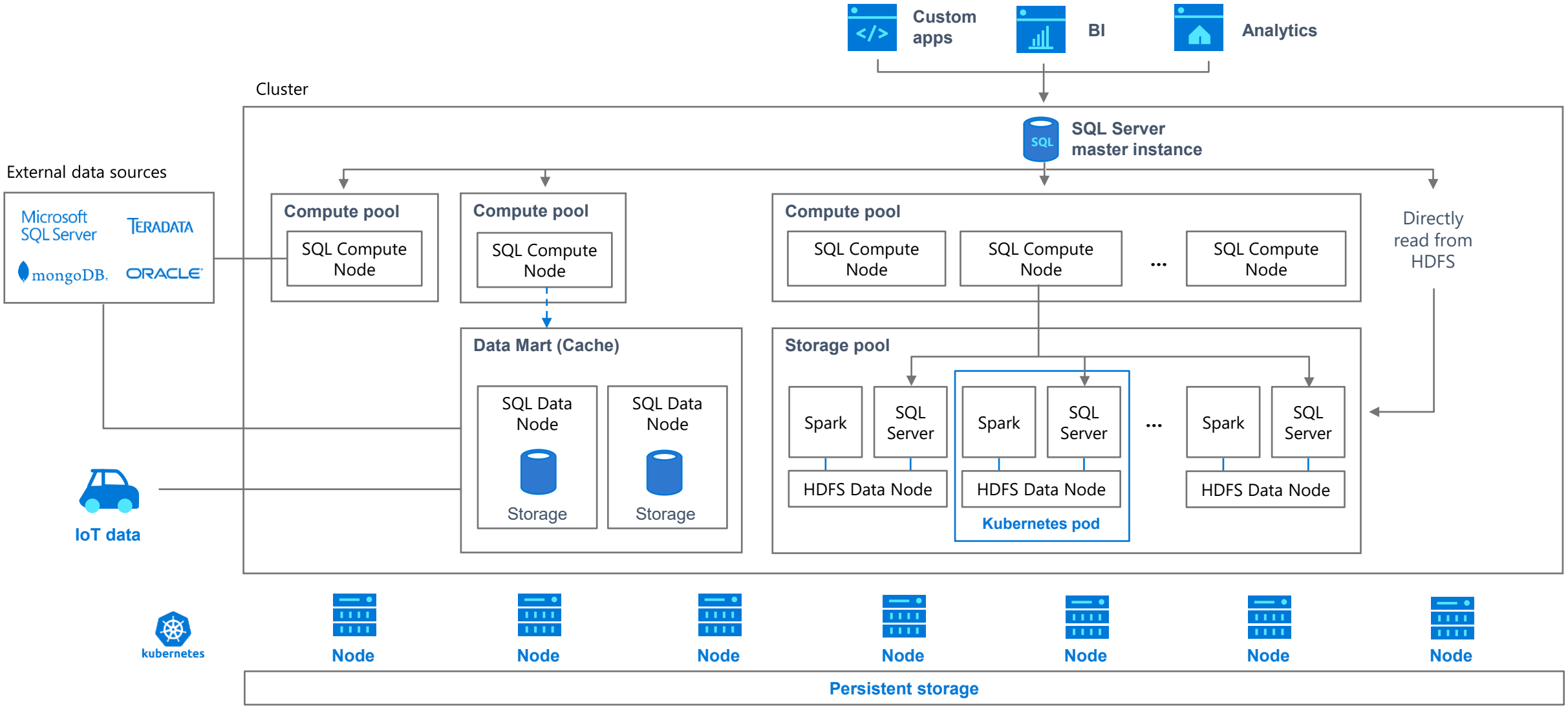
A confronto

External Tables	Linked Servers
Database Scoped	Instance Scoped
ODBC Drivers	OLEDB Providers
Read-only*	Read/write
Scale out queries with push-down	Single threaded queries with (limited) push-down
Failover with AG	Requires separate config from AG
Basic authentication	Basic and integrated authentication
Distributed Transactions not supported	Distributed Transactions supported
Better for OLAP workloads	Better for OLTP workloads

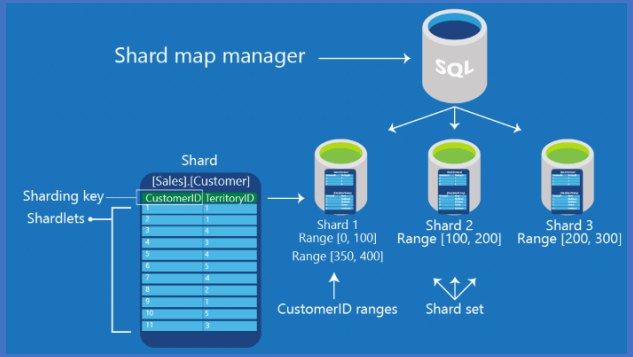
* Insert into HDFS allowed

SQL Server Big Data Cluster



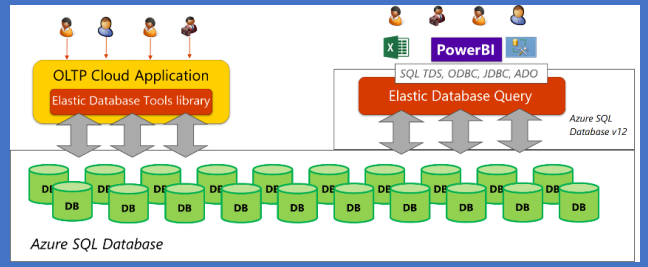
https://docs.microsoft.com/en-us/sql/big-data-cluster/big-data-cluster-overview

Scale-Out OLTP Azure SQL Database



- Elastic Database Client Library
  - Database Sharding & Data Dependent Routing
    - https://azure.microsoft.com/en-us/documentation/articles/sql-database-elastic-database-client-library
    - https://docs.microsoft.com/en-us/azure/sql-database/sql-database-elastic-scale-shard-map-management
  - Multi-Shard Queries (semantica UNION ALL)
    - https://docs.microsoft.com/en-us/azure/sql-database/sql-database-elastic-scale-multishard-querying

Scale-Out OLTP Azure SQL Database



- Elastic Database Queries
  - Supporto T-SQL per query distribuite
    - CREATE EXTERNAL DATA SOURCE
    - CREATE EXTERNAL TABLE
  - Partizionamento verticale: «cross-database queries»
    - TYPE = RDBMS
  - Partizionamento orizzontale: «Sharding»
    - TYPE = SHARD_MAP_MANAGER
  - https://azure.microsoft.com/en-us/documentation/articles/sql-database-elastic-query-overview

Scale-Out OLAP

- Azure Synapse Analytics
  - aka Azure SQL Data Warehouse
  - aka APS/PDW in Azure
  - PolyBase è nato qui, principalmente per integrazione HDFS

Altre soluzioni virtualizzazione OLAP

- Power BI
  - Limite memoria portato da 10GB a 400GB
  - 20-30% circa per esecuzione query
  - https://docs.microsoft.com/en-us/power-bi/power-bi-data-sources
- Azure Analysis Services
  - https://docs.microsoft.com/en-us/azure/analysis-services/analysis-services-datasource
- On-Premise Data Gateway
  - Può essere usato da Power BI o Azure Analysis Service
  - https://docs.microsoft.com/en-us/data-integration/gateway/service-gateway-onprem

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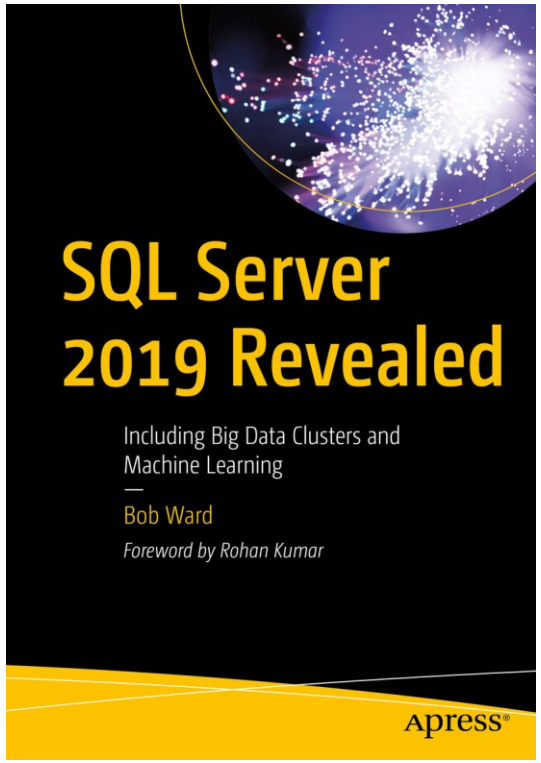
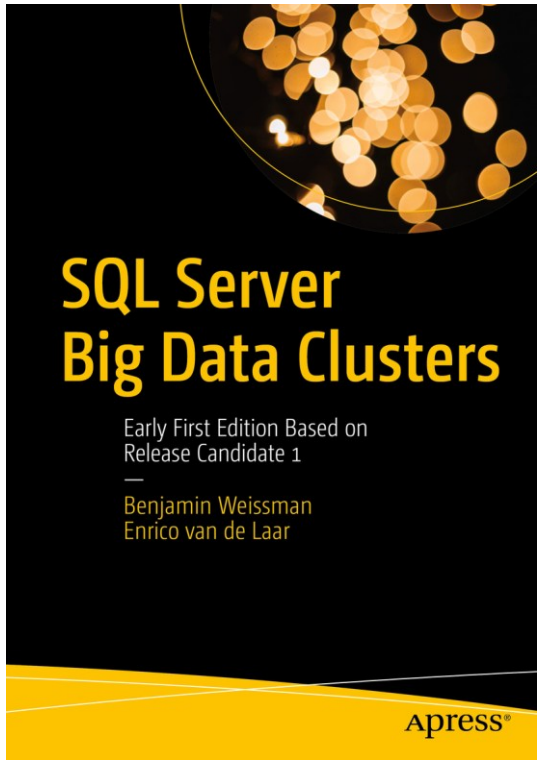
Risorse





Q&A

Contatti

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